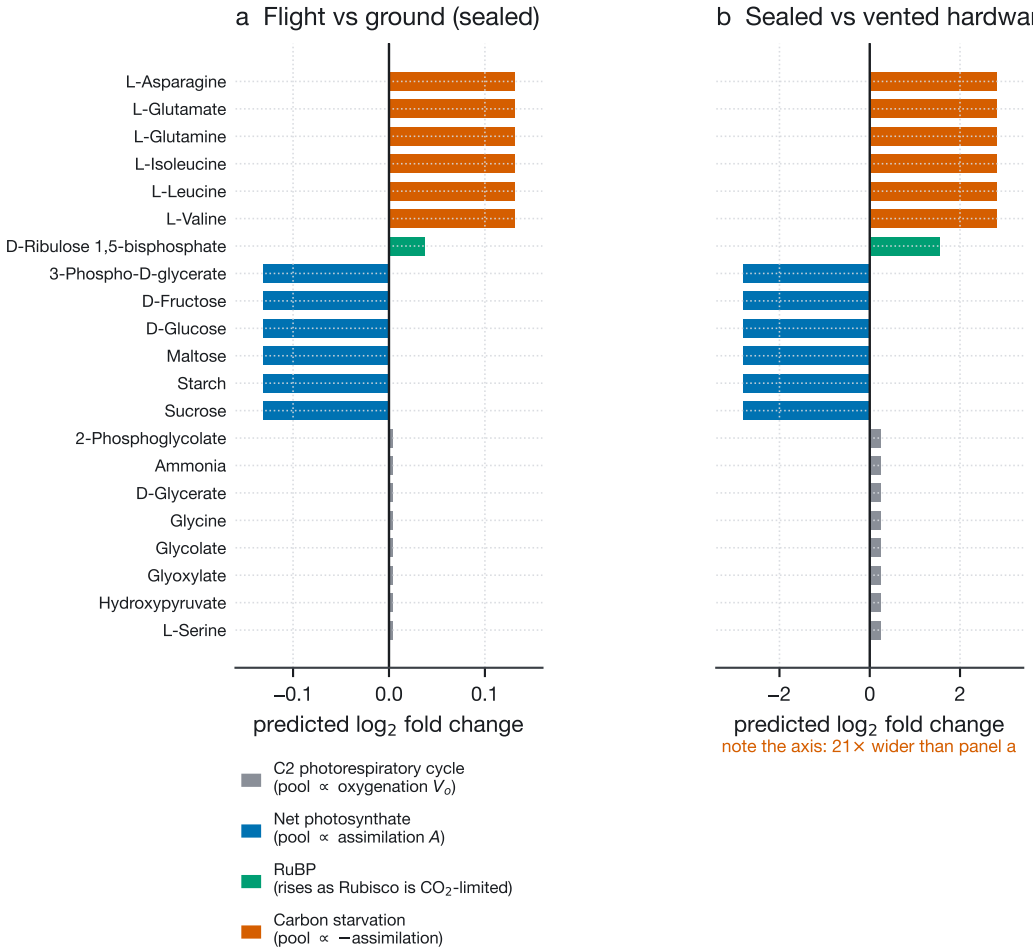
The metabolite layer had to be predicted — OSDR holds no plant metabolome



c Deliberately not predicted

TCA intermediates

the model does not resolve respiratory flux, and carbon limitation can deplete OR raise these pools

Fermentation products

fermentation needs hypoxia; a lit canister is releasing O_2

Glycine / serine ratio

both pools track the same flux, so the ratio cannot move — reporting it would be circular

The prediction has only four distinct values, because every pool in a block inherits the same flux ratio. That is what the model knows: it resolves fluxes, not individual pool kinetics.

EVERY VALUE ON THIS FIGURE IS MODEL OUTPUT. Nothing here was measured. No OSDR plant study has a metabolome.